

Response surface alignment measure

1 Set-up

AOD and ARI at the gridbox at longitude -171.5625 and latitude 51.875 (shown in green in Figure 1) are considered, for July.

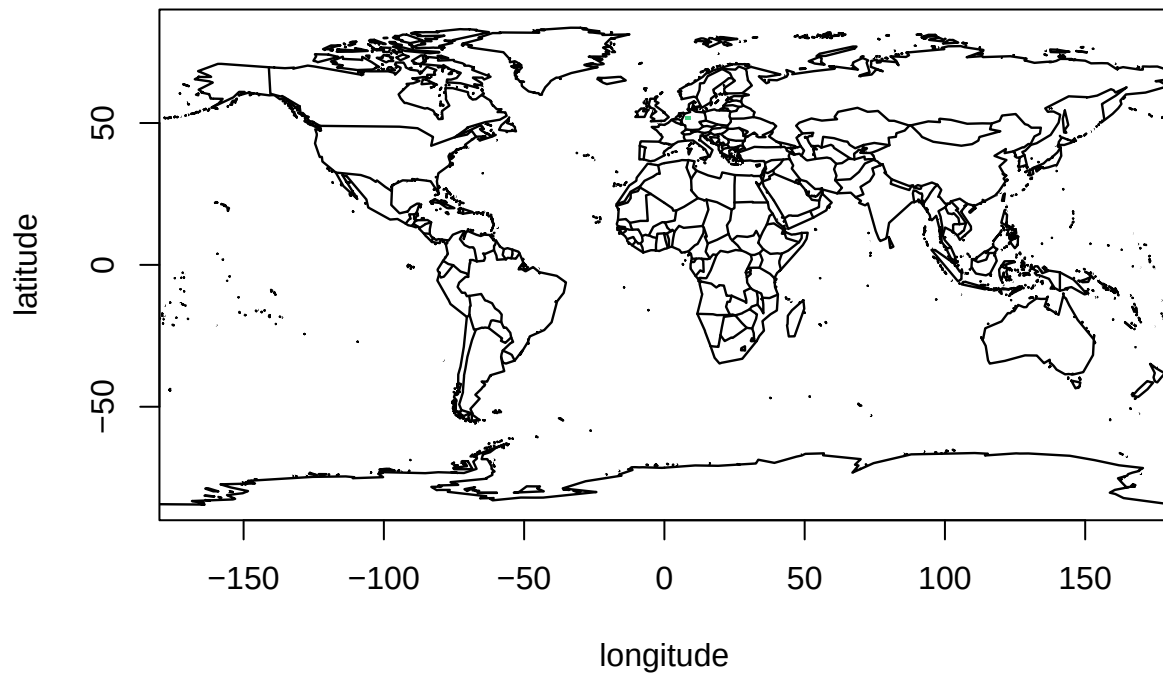


Figure 1: Selected gridboxes highlighted in green and blue.

2 Sensitivity analysis

GAM-based sensitivity analysis is performed for the two outputs. To reach 95% of the variance of the two outputs requires 7 and 9 inputs respectively (with main effects summing to 97% and 97%); see Table 1 and Figure 2..

Table 1: Post-SA main effects for the first (left) and second (right) outputs.

Input	Main effect (%)	Input	Main effect (%)
dry_dep_acc	42	prim_so4_diam	29
prim_so4_diam	20	dry_dep_so2	22
bvoc_soa	13	dry_dep_acc	12
kappa_oc	9	dry_dep_ait	7
sea_spray	6	anth_so2_eur	7
dry_dep_so2	4	anth_so2_r	6
anth_so2_eur	3	bc_ri	5
		anth_so2_chi	5
		anth_so2_asl	4

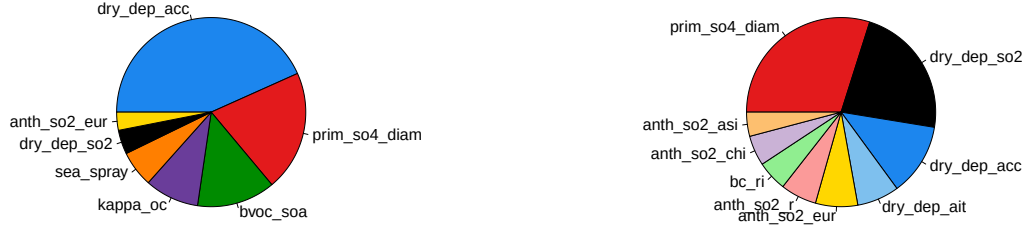


Figure 2: The variables with main effects summing to at least 95% for (left) output 1 and (right) output 2.

An emulator will be fitted for the union of inputs from Table 1, namely the 12 inputs:

1. dry_dep_acc
2. prim_so4_diam
3. bvoc_soa
4. kappa_oc
5. sea_spray
6. dry_dep_so2
7. anth_so2_eur
8. dry_dep_ait
9. anth_so2_r
10. bc_ri
11. anth_so2_chi
12. anth_so2_asl

3 Emulator fit and validation

To fit the emulator, we assume a linear mean function of the form

$$m(\mathbf{x}) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p.$$

The Gauss (squared exponential) kernel is used for the covariance function, maximum posterior mode estimation for the parameters estimates (trend, range, SD and the nugget term), and the **RobustGaSP** package to fit the emulators.

For validation of the emulators fitted, as well as calculation of the alignment measure, 100 new LHC points are generated and predictions made at these points. The mean width of the 95% CI's for these predictions are shown in Table 3, along with the mean width of the LOO CV prediction 95% CI's. Note that the range of output 1 is 0.6 and of output 2 is 2.6.

Table 2: Validation checks

Output	Prediction 95% CI mean width	LOO CV 95% CI mean width	% validated in LOO CV
1	0.13	0.13	95
2	0.66	0.66	94

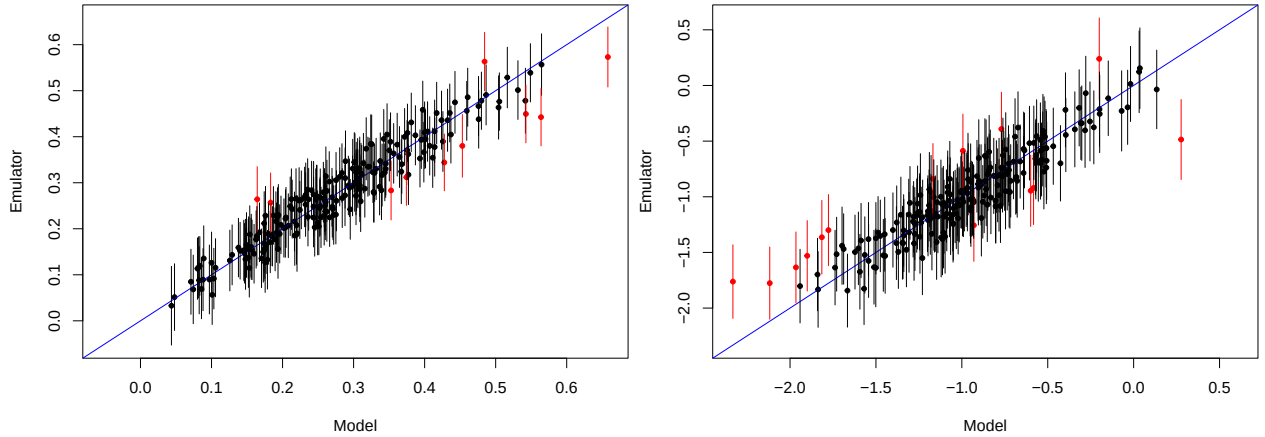


Figure 3: LOO CV validation plot for output 1 (left) and output 2 (right).

4 Check of by-hand calculations

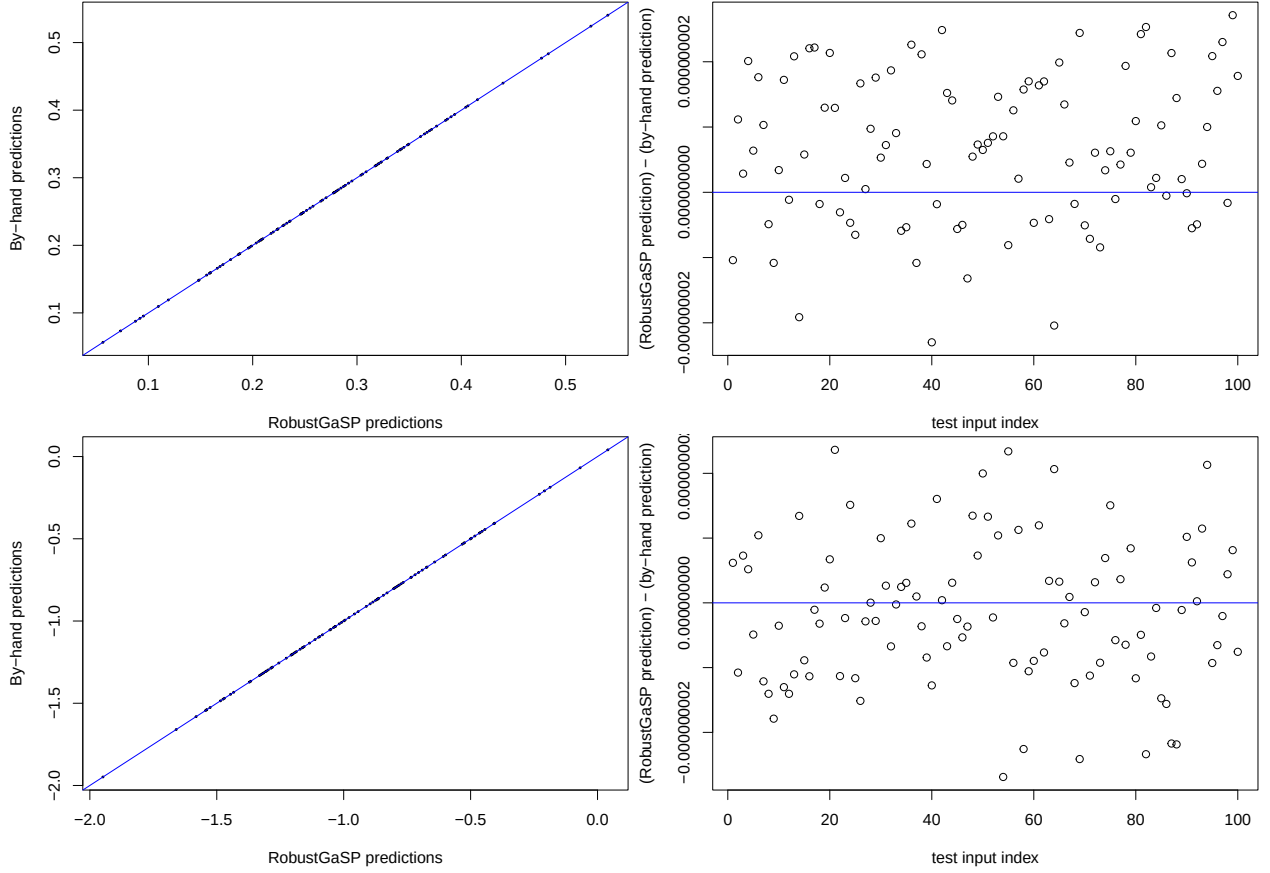


Figure 4: The predictions found by-hand match well with those made by the RobustGaSP package (left), with the latter minus the former found to be essentially zero (right). Top: Output 1. Bottom: Output 2.

Ahead of performing by-hand calculations of the partial derivatives at the testing points, by-hand calculations for the predictions made at these points are plotted against the predictions returned by the **RobustGaSP** package (left hand plots in Figure 3) and the difference between the two plotted against the test input index (right hand plots in Figure 3). Finally, for both we calculate

$$\frac{\sum_{i=1}^N \left| (\text{predictions from package})_i - (\text{predictions by-hand})_i \right|}{N},$$

where N is the number of test points to get 0 for output 1 and 0 for output 2.

5 Calculation of the alignment measure

Having calculated the emulated partial derivatives for use in the alignment measure, and normalising these, we calculate the AM, getting a value of 0.537.